

Integrations & API Requirements

External services, exposed interfaces, and the decisions we need from you

This document lists every third-party service the platform depends on, describes the API surface the backend exposes, and sets out the approvals and registrations required before each integration can be built.

Change in v1.1. Five suppliers confirmed: Razorpay, Cashfree Secure ID, Cloudflare R2, and Firebase, with SMS and email still open. One item needs your decision — **Cloudflare R2 cannot guarantee India-only data storage**, which affects the India-only commitment you gave in answer H3. See section 3.6.

Document	Integrations & API Requirements
Product	Networth Matrimony
Version	v1.1 — supersedes v1.0
Date	1 October 2026
Basis	Internal Developer Work Document (48 sections), client decisions of 29 September 2026, and supplier selections of 1 October 2026
Classification	Confidential — for client review

1. Purpose and scope

The platform has three parts: a mobile application for users, a web dashboard for administrators, and a backend API that both talk to. Almost every feature that carries commercial or legal weight — taking a payment, verifying an identity, sending a one-time password, releasing a contact number, storing a photograph — depends on a service outside our own code. This document is the complete list of those dependencies.

It answers three questions:

1. **What external services does the product need**, what does each one do, and what credentials or registrations must exist before we can use it?
2. **What API does our backend expose** to the mobile app and the admin dashboard, and what guarantees does it make?
3. **What do we need from you** — the approvals, accounts, and registrations that only you can authorise?

A note on status: “Configured” means the setting already exists in our local environment; “Partial” means a placeholder exists but the provider or credentials are not yet decided; “Not started” means nothing has been set up and the item is waiting on a decision below.

2. Integration summary

Service	What it is for	Spec ref.	Status
Razorpay Payments Confirmed 1 Oct	Collect unlock fees and the account setup fee; verify payment server-side; handle refunds.	Sections 4, 14, 29, 40	Confirmed
Cashfree Secure ID Identity verification Confirmed 1 Oct	Mandatory identity check at the final registration step. No full Aadhaar number is ever stored.	D9, GAP-9	Confirmed
SMS gateway One-time passwords	Deliver OTP codes at registration. SMS is used for OTP only, not general notifications.	Sections 6, 42; G1(b)	Open
Email Transactional email	Profile approval/rejection, payment, account status, and system announcements.	Section 42; G1(b)	Open
Firebase Cloud Messaging Push notifications Confirmed 1 Oct	All ordinary notifications, including the 2-hour unlock expiry reminder.	Section 42; C3, G1(b)	Confirmed
Cloudflare R2 Photo storage Confirmed 1 Oct	Store profile photos and serve short-lived links. Residency decision needed — see 3.6.	Sections 9, 41; H3	Action needed

Service	What it is for	Spec ref.	Status
Redis Cache, queue, rate limiting	Rate limiting, OTP storage, notification queues, and duplicate-request protection.	Section 41	Configured
PostgreSQL Primary database	The system of record for users, profiles, payments, unlocks, reports, and audit logs.	Section 35	Configured
Geocoding / maps Location & distance	Resolve cities to coordinates so discovery can sort by distance ("closest first").	E2, GAP-3	Not started
Scheduled jobs Background tasks	Unlock-expiry reminders, unlock expiry itself, and permanent deletion after the retention period.	C3, F6	Not started

3. Integration detail

3.1 Payments — Razorpay

Spec §4, §14, §29, §40

WHY THE PRODUCT NEEDS IT

The entire revenue model is a one-off card/UPI payment to unlock a profile's contact details for 24 hours. There is also a one-time ₹15 account setup fee (D9) collected after identity verification. Both must be taken through a payment gateway and verified on our server — the app is never trusted to declare a payment successful.

WHAT WE INTEGRATE

Create an order on the server with the price taken from our own pricing table; capture the gateway callback and webhook; verify the payment signature and the amount against the server-side order; record the payment; then create the 24-hour unlock. The webhook is made **idempotent**, so a repeated gateway message can never create a duplicate unlock. Refunds are admin-only and are limited to genuine technical faults (B5).

CREDENTIALS AND CONFIGURATION

Key ID, key secret, and webhook secret (held in our secret manager, never in the app). A GSTIN and an invoice-numbering scheme are needed for tax invoices.

PRICING SCHEDULE TO BE ENFORCED

Band	Base	GST (18%)	Total payable
Below ₹2 Cr	₹99.00	₹17.82	₹116.82
₹2 Cr – ₹5 Cr	₹249.00	₹44.82	₹293.82
₹5 Cr – ₹10 Cr	₹499.00	₹89.82	₹588.82
Above ₹10 Cr	₹999.00	₹179.82	₹1,178.82

Prices are stored in the database and are editable by a Super Admin; each change is audit-logged and existing payments keep their original amount.

CONFIRMED

Razorpay — Indian entity, full UPI/cards/netbanking support, GST-ready invoicing, and a webhook model that meets the idempotency requirement in Section 40.

DECISION NEEDED

Confirm the ₹15 setup fee is a flat total with no separate GST, whether it is refundable if identity verification fails, and whether it is charged once per account lifetime (see GAP-5).

3.2 Identity verification — Cashfree Secure ID

D9, GAP-9 · confirmed 1 Oct

WHY THE PRODUCT NEEDS IT

Identity verification is **mandatory** under decision D9. Until a user's identity is verified, they may neither view others nor be seen. This is also the strongest defence against the "fake profile" concerns raised in the specification.

WHAT WE INTEGRATE

An initiate-and-callback flow. The user proves identity through DigiLocker; we receive a transaction reference and a masked Aadhaar value. We store **only** that reference, the last four digits, and a consent record — never the full Aadhaar number.

CREDENTIALS AND CONFIGURATION

Cashfree environment (sandbox or production), client ID, and client secret, held in our secret manager.

COMPLIANCE NOTE

The DigiLocker flow is the one to use. It is consent-based, UIDAI- and DPDP-compliant, returns an already-masked number, and retains no copy of the document. Direct UIDAI access would require an AUA/KUA licence that most products do not hold — another reason Cashfree, as a licensed aggregator, is the right route. OCR and video KYC are also available but are not recommended here.

HOW WE RECORD IT

The verification *method* (DigiLocker) is stored separately from the *supplier* (Cashfree). The method is what varies between a DigiLocker flow and a direct UIDAI check; the supplier is a commercial detail that could change. Keeping them apart means a future change of aggregator is a small, safe change rather than a reinterpretation of past records.

STILL NEEDED FROM YOU

A Cashfree Secure ID account with sandbox and production credentials, and confirmation of the per-check pricing. Also confirm the platform may hold only the transaction reference and a masked last-4 digits, and never the full Aadhaar number (GAP-9).

3.3 SMS gateway — one-time passwords

§6, §42, G1(b)

WHY THE PRODUCT NEEDS IT

OTP by SMS is the only authentication channel at launch. Following decision G1(b), SMS is used for **OTP only**; all other notifications go by push and email. This is a deliberate cost decision.

WHAT WE INTEGRATE

Send OTP, receive delivery receipts, and enforce the OTP rules already defined: a 5-attempt limit, a 60-second resend cooldown, and a 5-minute validity window.

WHERE THINGS STAND

Still open. You are looking for a cheaper alternative to MSG91, with MSG91 as the fallback if nothing better is found. We have therefore left the configuration provider-neutral, so that changing provider is a configuration change and not a code change. The integration work is unaffected either way.

CREDENTIALS AND CONFIGURATION

API key, **plus** the values that Indian SMS delivery legally requires: a registered sender ID, an approved message template, and the DLT entity ID. These are configured and waiting.

COMPLIANCE NOTE

India's TRAI regulations require every transactional SMS sender, template, and entity to be registered on the DLT platform before messages will deliver. This registration is done by the business, not by us, and it takes time — it should begin early, because it is required *whichever* provider you choose.

NO DECISION NEEDED YET — ONE ACTION

Provider choice is not blocking. Please begin DLT registration (entity, sender ID, and OTP template) now, since it is required regardless of the outcome and until it completes, no OTP will reach users.

3.4 Email — transactional

§42, G1(b)

WHY THE PRODUCT NEEDS IT

Email carries profile approval and rejection notices, payment receipts, account-status messages, and system announcements — including the account data-export and deletion confirmations required by the privacy work (H3).

WHAT WE INTEGRATE

A transactional-send API with delivery tracking, driven by the same event list as push. We also need a sending domain with SPF, DKIM, and DMARC configured, so messages are not treated as spam.

WHERE THINGS STAND

Still open, and being assessed alongside SMS. Email carries no residency obligation of its own — the messages we send contain no personal data beyond what the user has already given us — so the provider choice can be made on price and deliverability alone. If you settle SMS and email with the same supplier, that is usually the cheapest route.

CREDENTIALS AND CONFIGURATION

Provider API key plus the verified sending domain and a from-address. The domain also needs SPF, DKIM, and DMARC records so messages are not treated as spam.

3.5 Push notifications — Firebase Cloud Messaging

§42, C3, G1(b)

WHY THE PRODUCT NEEDS IT

Push is the primary channel for the notification events in Section 42: profile approved or rejected, payment successful, unlock nearing expiry (2 hours before, per C3), unlock expired, account suspended, and announcements.

WHAT WE INTEGRATE

Device-token registration when the app installs or the token refreshes, then server-side sends against the event list above.

CONFIRMED

Firebase, as recommended. The supported mechanism is the FCM **HTTP v1** API using a service-account credential; the legacy server key is deprecated, and we have removed it from our configuration so no deployment can rely on a credential that no longer works.

CREDENTIALS AND CONFIGURATION

A Firebase project with a service-account credential, plus device-token registration in the app.

COMPLIANCE NOTE

Push delivery passes through Google's infrastructure. The notification payload should therefore carry no sensitive profile data — only a short message and an identifier the app resolves after it authenticates. This is worth stating plainly in the Privacy Policy: it is the one part of the stack whose data path is not fully within India.

3.6 Photo storage — Cloudflare R2

\$9, \$41, H3 · confirmed 1 Oct

WHY THE PRODUCT NEEDS IT

Profile photos are uploaded, moderated, and displayed. Photos are visible before payment, which makes them a scraping target.

WHAT WE INTEGRATE

Upload via short-lived signed URLs, server-side validation of type, dimensions, and size, thumbnail generation, and display through short-lived signed links (5 minutes by default) so images cannot be harvested. R2 speaks the same S3 interface we already use, so this is a configuration change rather than new code.

CREDENTIALS AND CONFIGURATION

Account-scoped endpoint, bucket, access key, and secret. Note that R2 is addressed by *account* rather than by region, so the “region” setting is simply `auto`.

WHY YOU CHOSE IT

Legitimate cost benefit. R2 has no egress charges, and photos are the one part of this system that is served repeatedly, so delivery cost is real rather than theoretical.

DECISION NEEDED — THIS ONE MATTERS

Cloudflare R2 cannot guarantee India-only data storage. R2’s hard residency controls, which Cloudflare calls *jurisdictions*, are only:

`eu` (European Union) · `us` (United States) · `fedramp` · `default` (no restriction)

There is **no India jurisdiction**. The only Asia-Pacific control is a *location hint* (`apac`), and Cloudflare documents these as best-effort placement — explicitly **not** a residency guarantee. Hints are also applied only when a bucket is first created.

In answer H3 you committed the platform to India-only storage. With R2, user photographs cannot be asserted to be in India. For a platform holding religion, caste, income, family details and contact numbers, that is a real weakening of that commitment, so it should be a decision rather than a side effect. Internally we track this as **GAP-11**.

Three ways forward — please choose one:

Option	What it means	Trade-off
1	Accept it. H3 becomes “Asia-Pacific, best-effort”, and the Privacy Policy says so plainly.	Cheapest, keeps the R2 saving. We do not recommend this: it quietly changes a commitment you already made in writing.
2	Hybrid. Keep an India-region bucket (AWS S3 <code>ap-south-1</code>) for user photos and identity documents, and use R2 for non-sensitive assets such as app imagery.	Two vendors to manage, but the sensitive data stays in India and the R2 saving is largely kept. Our recommendation.
3	Keep everything in India and drop R2.	Simplest, and the only option that leaves H3 exactly as you answered it. Gives up the egress saving.

3.7 Redis and PostgreSQL — internal infrastructure

\$35, \$41

WHY THE PRODUCT NEEDS THEM

These are not external vendors but they are infrastructure dependencies. PostgreSQL is the system of record. Redis backs rate limiting, OTP storage, background queues, and the duplicate-payment protection from B3.

CREDENTIALS AND CONFIGURATION

Connection strings. Both are configured, including a 30-table database schema and a seed of reference data (net-worth bands, pricing, and permission roles).

3.8 Geocoding and distance — location services

E2, GAP-3

WHY THE PRODUCT NEEDS IT

Decision E2 asks for discovery results ordered “newest and closest”. The word *closest* implies distance between users. This is only possible if cities are resolved to coordinates and distance is computed — which needs a geocoding source or a maps provider.

WHAT WE INTEGRATE

Either a geocoding API to turn the stored city into coordinates, or a pre-loaded India city-coordinate database, after which distance is calculated inside our own database (no per-request external calls).

COMPLIANCE NOTE

Preferably avoid sending user locations to a third party at all. A static city-coordinate table keeps the dependency small and the data home.

OUR RECOMMENDATION

A bundled India city-coordinate dataset with in-database distance sorting. A hosted geocoder (for example Google Maps Platform) only if precise address-level distance is later required.

DECISION NEEDED

Confirm the discovery sort order (GAP-3): newest first with distance as a tie-breaker, or a genuine blend of the two. This decides whether a maps budget is needed at all.

3.9 Scheduled jobs — reminders and data purge

C3, F6

WHY THE PRODUCT NEEDS IT

Three things happen on a timer, not in response to a request: the 2-hour unlock-expiry reminder (C3), the expiry of the 24-hour unlock itself, and the permanent erasure of a deleted account after six months (F6).

WHAT WE INTEGRATE

A background worker (not an external vendor) driven by the Redis queue, plus a scheduler that runs the periodic sweeps.

CREDENTIALS AND CONFIGURATION

Only the notification and database credentials already listed; no new vendor.

4. API surface we expose

The backend exposes a versioned REST API consumed by two separate permission domains: the mobile application and the admin dashboard. The groups below follow Section 37 of the specification.

API group	Responsibility
/auth/*	Registration, OTP verification, sessions, token refresh, logout.
/users/*	Account details, settings, notification preferences, account deletion.
/profile/*	Profile creation and editing, in saveable sections.
/photos/*	Upload, reorder, delete, and mark the primary photo.
/partner-preference/*	Saved discovery preferences and reset/save actions.
/discovery/*	Same-category discovery, filters, and pagination.
/profiles/*	The locked preview and the unlocked full profile with contact.
/payments/*	Order creation, verification, webhook handling, history.
/unlocks/*	Active unlocks, remaining time, and expiry.
/reports/*	Submitting and tracking profile reports.
/blocks/*	Blocking, unblocking, and the blocked-user list.
/notifications/*	Reading notifications and managing delivery settings.
/admin/*	All dashboard operations: users, approvals, payments, pricing, reports, audit.

Behavioural guarantees

Every protected endpoint enforces the following, as required by Sections 37–40:

- Authenticate every protected request, and authorise ownership of the resource.
- Apply the net-worth category filter on the server; a request for the opposite category is rejected.
- Return only permitted fields — the pre-payment profile response is a closed schema, so a hidden field cannot leak under another name, nested object, or metadata.
- Verify an active unlock before returning the full profile or contact details.
- Validate all pagination and filter inputs, and cap page sizes.
- Rate-limit sensitive endpoints (OTP, login, payment, unlock).
- Record security-sensitive operations in an audit log.
- Treat payment webhooks as untrusted: verify the signature, verify the amount against the server-side order, and de-duplicate repeated events.

5. Compliance and data residency

Several requirements affect every integration above and are not optional:

- **India-only data residency** (H3). All user data, including photographs, is intended to be stored and processed in India. **One exception is now open:** Cloudflare R2 cannot guarantee India residency, and Firebase push notifications pass through Google's infrastructure. Please see section 3.6 for the decision we need on storage.
- **Data protection.** The Digital Personal Data Protection Act applies. Consent is captured explicitly, only necessary data is collected, and users can export and delete their data.
- **Aadhaar minimisation.** The full Aadhaar number is never stored — only the verification reference and masked last four digits (GAP-9). Cashfree's DigiLocker flow returns an already-masked value and retains no document, which matches this rule without loosening it.
- **Retention.** Deleted accounts are retained for six months and then irreversibly erased (F6).
- **Legal documents.** We are drafting the Privacy Policy and Terms of Service for professional review (H1(b)). They must describe the storage and push-notification positions accurately, which is why the section 3.6 decision matters for the wording.

6. What we need from you

The following items are outside our control and block the corresponding integration. The right-hand column is our recommended sequencing, not a deadline you have set.

Item	What we need	Blocks	Priority
Cloudflare R2 vs India residency	Choose one of the three options in section 3.6. R2 cannot guarantee India-only storage, which conflicts with your H3 answer.	Photo storage, Privacy Policy wording	Critical
₹15 setup-fee rules	Confirm GST treatment, refundability, and whether it is once per account (GAP-5).	Setup-fee payment	Critical
Cashfree Secure ID account	Sandbox and production credentials, and confirmation of per-check pricing.	Mandatory identity verification	Critical
DLT registration	Register entity, sender ID, and OTP template for transactional SMS. Required whichever provider you choose.	All OTP delivery	Critical
Razorpay account	Merchant account, API keys, webhook secret, GSTIN, and invoice details.	All payments and unlocks	High
Firebase service account	Create a Firebase project and supply the HTTP v1 service-account credential.	All push notifications	High
Email sending domain	Provide the domain and approve DNS records for SPF, DKIM, and DMARC.	All transactional email	High

Item	What we need	Blocks	Priority
SMS and email provider	Open. MSG91 is the fallback. No architectural impact, so this can be settled late. Internally tracked as GAP-12.	Notification delivery	Medium
Discovery sort order	Confirm “newest and closest” precedence (GAP-3).	Discovery ranking, maps budget	Medium
Family photo rules	Required or optional, which is the cover image, visible before payment (GAP-4).	Photo moderation, screening	Medium
Legal review	Professional review and sign-off of the Privacy Policy and Terms.	App-store submission	Medium
Who may delete a user	Confirm the role permitted to delete accounts (GAP-6).	Admin deletion, audit rules	Low

7. Indicative cost drivers

Figures below are placeholders to show *which* services bill per use. Exact rates depend on the providers and volumes you approve; we will confirm them before any is selected.

Service	Charged by	Notes
SMS (OTP)	Per message delivered	Small per-message cost; volume tracks registrations and resends.
Identity verification	Per successful check	Charged per verified user via Cashfree Secure ID; likely the largest variable cost.
Payment gateway	Percentage of transaction	Standard Indian gateway rate applies to each unlock and setup fee.
Push notifications	Free at our volume	Firebase Cloud Messaging has no per-message charge at this scale.
Photo storage	Per GB stored	Cloudflare R2 storage is low-cost.
Photo delivery / bandwidth	Free on R2 · per GB on AWS	Material. Photos are the only repeatedly served asset, so this is the largest storage variable cost and the main financial argument for R2. If you choose the hybrid option in 3.6, photos served from R2 cost nothing to deliver.
Geocoding / maps	Per request (if hosted)	Avoidable entirely with a bundled city-coordinate dataset.

Service	Charged by	Notes
Email	Per message sent	Negligible at launch volumes; AWS SES rates apply.

8. Assumptions and next steps

Assumptions

- Razorpay, Cashfree Secure ID, Cloudflare R2, and Firebase are treated as settled. SMS and email remain open with MSG91 as the fallback.
- The database schema records the identity verification method and the supplier separately, so a later change of supplier does not alter historical records.
- All four net-worth bands and their prices are provisional bootstrap values; the database is authoritative and prices are editable by a Super Admin.
- Prices, categories, and role permissions are stored as data, so adding a band later is a data change, not a rebuild.

Next steps

1. You choose one of the three storage options in section 3.6. This is the only item that changes what we build rather than merely when.
2. You return the accounts and registrations on the critical items in section 6 — Razorpay, Cashfree, and DLT.
3. We confirm exact rates for the variable-cost services, once the storage option is settled.
4. Payments and identity verification are built first, as they sit on the critical path. Storage, SMS, email, and push follow.

Networth Matrimony — Integrations & API Requirements · v1.1 · 1 October 2026. Confidential, for client review. Prepared from the Internal Developer Work Document, the client decisions recorded on 29 September 2026, and the supplier selections confirmed on 1 October 2026. Supersedes v1.0, which recommended S3 in ap-south-1 and left the identity supplier open. Open items referenced as GAP-n are detailed in the internal decisions log.